

Gas Tests

Here are the field tests for five common gases: carbon dioxide, hydrogen, oxygen, chlorine, and ammonia.

Hydrogen Gas (H₂)

- Test: Squeaky pop test
- Method: Introduce a **lit splint** into a sample of the gas. Result: A
- **squeaky pop** sound.

Oxygen Gas (O₂)

- Test: Glowing splint test
- Method: Introduce a **glowing splint** into a sample of the gas.
- Result: The splint **re-lights**.

Chlorine Gas (Cl₂)

- Test: Damp blue litmus paper
- Method: Expose **damp blue litmus paper** to the gas.
- Observations:
 - The litmus paper turns **red** initially.
 - The litmus paper then **bleaches**.

Ammonia Gas (NH₃)

- Test: Damp red litmus paper
- Method: Expose **damp red litmus paper** to the gas. Result: The
- litmus paper turns **blue**.

Carbon Dioxide (CO₂)

- Test: Lime water test
- Method: **Bubble** the gas through **lime water**.
- Result: The lime water turns **cloudy**, forming a **white precipitate** (calcium carbonate).

Halide Ion Tests

Test for chloride, bromide, and iodide ions using the silver nitrate test. Note that fluoride ions are not tested as they are all soluble.

Method

1. Add **nitric acid** (HNO₃) to the sample to remove any carbonate ions.
2. Add **silver nitrate** (AgNO₃).
3. Observe the precipitate color.

Results

Halide Ion	Precipitate Color
Chloride (Cl ⁻)	White
Bromide (Br ⁻)	Cream
Iodide (I ⁻)	Yellow

Flame Tests

Five GCSE flame colors to remember:

- Lithium (Li⁺)
- Sodium (Na⁺)
- Potassium (K⁺)
- Calcium (Ca²⁺)
- Copper (Cu²⁺)

Method

1. Use a **nichrome wire loop**.

Nichrome wire is inert.

2. Clean the loop by dipping it in **hydrochloric acid**. This removes any previous chemicals and helps the solid stick to the wire.
3. Dip the loop into the solid compound being tested.
4. Place the loop into the flame and observe the color.

Results

Ion	Flame Color
Lithium (Li^+)	Red
Sodium (Na^+)	Yellow
Potassium (K^+)	Lilac
Calcium (Ca^{2+})	Orange-Red
Copper (Cu^{2+})	Blue-Green

Flame tests are used to identify metal ions based on the color they produce when heated in a flame. Not all metal ions produce distinct flame colors, so other tests are also necessary.

- **Sodium (Na)**: Leftover sodium can produce a yellow color, which can interfere with other flame tests.
- **Potassium (K)**: Produces a lilac (pale purple/pink) color. This color can be difficult to see.
- **Calcium (Ca)**: Yields a brick red color. It can sometimes appear orange, but it's specifically a brick red, like a mix of yellow and red.
- **Copper (Cu)**: Results in a green flame when copper sulfate is heated.

Sodium Hydroxide Precipitate Tests

These tests involve adding sodium hydroxide (NaOH) to solutions containing metal ions to observe the color of the precipitate formed.

- **Iron (II) or Fe^{2+}** : Produces a murky green solid precipitate. This is iron (II) hydroxide.
- **Copper (II) or Cu^{2+}** : Forms a murky blue precipitate. This is copper (II) hydroxide.
- **Iron (III) or Fe^{3+}** : Creates a horrible brown precipitate, which is iron (III) hydroxide.

Tests for Complex Ions

Sulfate Ion Test (SO_4^{2-})

1. **Remove Carbonate Ions**: Add a small amount of hydrochloric acid (HCl) to remove any carbonate ions that may interfere with the test.
2. **Add Barium Chloride (BaCl_2)**: If sulfate is present, a white precipitate of barium sulfate (BaSO_4) will form.

Ammonium Ion Test (NH_4^+)

1. **Add Sodium Hydroxide (NaOH)**: Add a quantity of sodium hydroxide to the sample.
2. **Warm the Mixture**: Gently heat the solution.
3. **Test for Ammonia Gas (NH_3)**: Use damp red litmus paper. If ammonia gas is present, the litmus paper will turn blue.

Carbonate Ion Test

Add any acid to the sample. If carbonate ions are present, bubbles of carbon dioxide (CO_2) will be produced.

Summary of Ion Tests

Ion	Test	Result
K^{+}	Flame Test	Lilac flame
Ca^{2+}	Flame Test	Brick red flame
Cu^{2+}	Flame Test	Green flame
Fe^{2+}	Sodium Hydroxide	Green precipitate
Cu^{2+}	Sodium Hydroxide	Blue precipitate
Fe^{3+}	Sodium Hydroxide	Brown precipitate
SO_4^{2-}	Hydrochloric Acid, then Barium Chloride	White precipitate
NH_4^{+}	Sodium Hydroxide, then warm. Test gas with litmus paper	Damp red litmus paper turns blue
CO_3^{2-}	Add Acid	Bubbles of CO_2 are produced

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